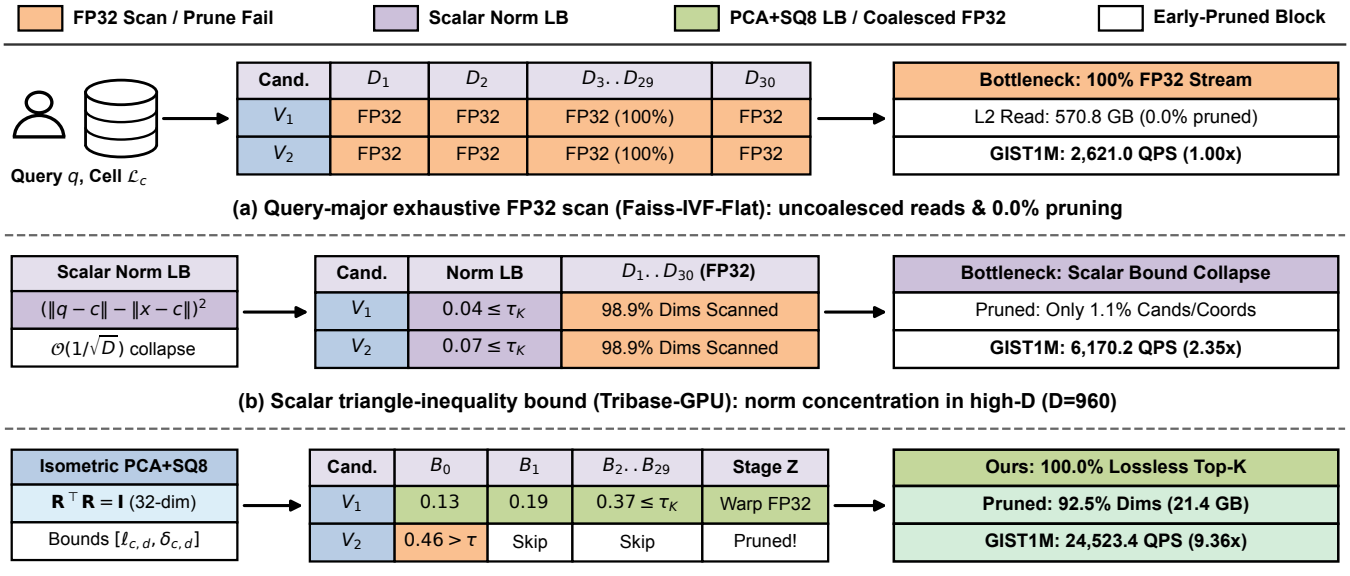

FP32 Scan / Prune Fail
Scalar Norm LB
PCA+SQ8 LB / Coalesced FP32
Early-Pruned Block
Query q, Cell $\mathcal{L}_c$
Cand. | D_1 | D_2 | $D_3..D_{29}$ | D_{30}
V_1 | FP32 | FP32 | FP32 (100%) | FP32
V_2 | FP32 | FP32 | FP32 (100%) | FP32
Bottleneck: 100% FP32 Stream
L2 Read: 570.8 GB (0.0% pruned)
GIST1M: 2,621.0 QPS (1.00x)
(a) Query-major exhaustive FP32 scan (Faiss-IVF-Flat): uncoalesced reads & 0.0% pruning
Scalar Norm LB
$(\|q-c\| - \|x-c\|)^2$
$\mathcal{O}(1/\sqrt{D})$ collapse
Cand. | Norm LB | $D_1..D_{30}$ (FP32)
V_1 | $0.04 \le \tau_K$ | 98.9% Dims Scanned
V_2 | $0.07 \le \tau_K$ | 98.9% Dims Scanned
Bottleneck: Scalar Bound Collapse
Pruned: Only 1.1% Cands/Coords
GIST1M: 6,170.2 QPS (2.35x)
(b) Scalar triangle-inequality bound (Tribase-GPU): norm concentration in high-D (D=960)
Isometric PCA+SQ8
$\mathbf{R}^\top\mathbf{R} = \mathbf{I}$ (32-dim)
Bounds $[\ell_{c,d}, \delta_{c,d}]$
Cand. | B_0 | B_1 | $B_2..B_{29}$ | Stage Z
V_1 | 0.13 | 0.19 | $0.37 \le \tau_K$ | Warp FP32
V_2 | $0.46 > \tau$ | Skip | Skip | Pruned!
Ours: 100.0% Lossless Top-K
Pruned: 92.5% Dims (21.4 GB)
GIST1M: 24,523.4 QPS (9.36x)
(c) TRIDENT cluster-major progressive dimension-block pruning (PCA + per-cluster SQ8 LB)